

Assemble a Six Ingredient Salad SOP

One episode builds one salad and splits it across two plates. Greens and five toppings go into the mixing bowl in a fixed order, the salad is dressed from a squeeze bottle, tossed with two servers, and portioned onto two plates. The tray is empty, the tongs and both servers are back in their rest, and the bottle is back on its spot when the episode ends.

The fill order never changes. Greens, then topping 1 through topping 5 in tray order.

Three tools do three jobs and each one lives in a rest between uses. The tongs load every ingredient. The bottle dresses. The two servers toss and plate. No tool is ever held while another tool is being used, and no tool is ever set down outside its rest.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the ingredient tray on the left, the utensil rest, the mixing bowl in the middle, the dressing bottle, and both plates on the right.
3. Both arms are at home with grippers open.
4. The tabletop is clear of anything but the zones listed below.
5. Nothing obstructs the paths between the tray, the mixing bowl, and the plates.

Materials checklist

1. The ingredient tray sits against the left edge and holds six open containers, none of them overlapping or touching: the greens tub in the back row, and five topping bowls in the front row in a straight line.
2. The greens tub holds one episode's worth of loose leaves, filled to below its rim so the tongs can close inside the tub without leaves spilling over the side.
3. Each of the five topping bowls holds one episode's portion of a single topping. Nothing is mixed between topping bowls and no topping bowl is filled above its rim.
4. The topping bowls are numbered 1 to 5 in tray order, topping bowl 1 nearest the mixing bowl and topping bowl 5 furthest from it. That numbering is fixed at setup and does not change mid episode.
5. The utensil rest sits between the tray and the mixing bowl and holds three tools standing in their own slots, handles up and clear of the rest so a gripper can take each from above: the tongs, the left server, and the right server. All three are clean and dry.

6. The mixing bowl sits in the middle of the tabletop on its non-slip ring, empty and dry. The ring holds the mixing bowl still, so no gripper ever has to.
7. The dressing bottle stands upright on its marked spot to the right of the mixing bowl, within reach of the right arm. It is a squeeze bottle with an open spout, the cap already off the bottle and set aside off the tabletop, and it holds enough dressing that it never has to be shaken or squeeze it dry.
8. Both plates sit against the right edge, empty and right side up, side by side and not touching. The plate nearer the mixing bowl is the near plate and the other is the far plate.

Workspace layout

- **Ingredient tray:** on the tabletop against the left edge, within reach of the left arm. It holds the greens tub and the five topping bowls. Nothing is ever set down on it.
- **Utensil rest:** the slotted block between the tray and the mixing bowl, within reach of both arms. The tongs and both servers live here whenever they are not in a gripper.
- **Mixing bowl:** on its non-slip ring in the middle of the tabletop, reachable by both arms. Every ingredient lands here and the whole toss happens here.
- **Dressing bottle:** upright on its marked spot to the right of the mixing bowl, within reach of the right arm.
- **Plates:** on the tabletop against the right edge, within reach of the right arm.

Naming: the big bowl in the middle is always called the **mixing bowl**. The five small ones on the tray are always called **topping bowls**, by number. Nothing in this document is just "the bowl".

Each arm owns the side it sits on. The left arm works the tray and the tongs. The right arm works the bottle and the plates. Both arms reach the rest and the mixing bowl.

Vocabulary

Arm assignments

- **The left gripper owns the tray and the tongs.** It loads all six ingredients into the mixing bowl. It never touches the bottle and never reaches a plate.
- **The right gripper owns the dressing bottle and the two plates.** It never picks anything off the tray and never takes the tongs.
- **Both grippers take a server.** The left gripper takes the left server, the right gripper takes the right server, and they keep those servers for the toss and for the plating.
- **Plating is the one two-arm job on the right side.** Both servers carry a portion to the plates together, so the left arm does reach the plates while it is holding its server, and only then.

- **Nothing is passed hand to hand.** A tool that changes arms does so by going back to its slot in the rest first.
- A gripper opens only over the mixing bowl, over a plate, or into a slot in the rest. Never open a gripper over the tray, over a topping bowl, or over open tabletop.

Handling the tools

Each tool has one job and one home. The rest is the home for all three.

- **Tongs:** the left gripper takes them from above, across both handles near their top ends. Closing the gripper closes the tongs. The tongs load every ingredient and nothing else.
- **Bottle:** the right gripper takes it around the body, below the neck and above the base. Squeezing the gripper squeezes the bottle.
- **Servers:** each gripper takes its own server from above by the top of the handle, one server per gripper.
- **A tool goes back to its rest before the next tool is picked up.** Never hold two tools at once and never lay a tool down on the tabletop, on the tray, or across the mixing bowl rim.
- **Change jobs through the rest, never in the air.** No gripper shifts its hold along a handle while the tool is off its rest.
- Carry every tool level and clear of the rim of anything it passes over.

The fill order

- **Tray order, not reach order.** Greens first, then topping bowl 1, 2, 3, 4, 5 in that order, whatever is in them.
- The greens always go in first. They are the base and every topping lands on top of them.
- **One ingredient at a time.** Never take from two containers on one trip and never combine two toppings in the tongs.
- Empty each topping bowl in one load. If some is left behind, take a second load with the same tongs before returning them to the rest.

Handling standard

- Bring the tongs down inside the rim of the mixing bowl before opening them, so nothing falls in from height and bounces back out.
- Bring the tongs down inside the rim of a container before closing them, so nothing is dragged over the side.
- **Never grip, steady, lift, or slide the mixing bowl.** The ring holds it. If the mixing bowl has moved off its ring, stop and reseal it before continuing.
- **Never grip an ingredient directly with a gripper.** Everything that goes in the mixing bowl gets there through the tongs or the servers.

- **Never press or pack the salad down.** Nothing in this episode presses on the contents of the mixing bowl or of a plate.
- Spilled leaves or toppings on the tabletop are left where they fall. Do not stop to collect them.

Steps

Step 1: Load the greens

Goal: all of the greens are in the mixing bowl and the tongs are back in their rest.

- The **left gripper** takes the tongs from the rest and lifts them straight up out of the slot.
- Bring the tongs down into the greens tub, close them on a load of leaves, and lift straight up.
- Carry the tongs level to the mixing bowl, lower them inside the rim, and open them to release the load.
- Repeat for a second load. The tongs stay in the gripper between the two loads.
- Take further loads the same way until the greens tub is empty.
- Carry the tongs back to the rest, lower them into their slot until seated, and release.

Check: the greens tub is empty, the greens sit in the mixing bowl, and the tongs are seated in their slot. If leaves are hanging over the mixing bowl rim, push them in with the tongs before returning them.

Step 2: Load the five toppings

Goal: all five topping bowls are empty, their contents are in the mixing bowl on top of the greens, and the tongs are back in their rest.

Work one topping bowl at a time in tray order: topping bowl 1, then 2, 3, 4, and 5.

2.1 The load cycle

Run this cycle once per topping bowl, in full, five times.

- The **left gripper** takes the tongs from the rest and lifts them straight up out of the slot.
- Bring the tongs down inside the topping bowl, close them on the whole portion, and lift straight up.
- Carry the tongs level to the mixing bowl, lower them inside the rim, and open them to release the load.
- If anything is left in the topping bowl, go straight back for a second load with the same tongs. Do not return them to the rest part way through a topping bowl.
- Once the topping bowl is empty, carry the tongs back to the rest, lower them into their slot until seated, and release.

Do not skip a topping bowl because it looks nearly empty, and do not go back to an earlier topping bowl once you have moved on.

Check after each topping bowl: the topping bowl is empty, its contents are in the mixing bowl, and the tongs are seated in their slot before the next topping bowl starts.

Expected state: the tray holds six empty containers. The mixing bowl holds the greens and all five toppings, undressed and untossed. All three tools are in the rest.

Step 3: Dress the salad

Goal: the dressing is on the salad and the bottle is upright on its marked spot.

- The **right gripper** takes the bottle around the body and lifts it straight up off its spot.
- Carry it level to the mixing bowl and bring the spout over the middle of the mixing bowl, inside the rim.
- Tip the bottle to pour and squeeze steadily for a count of 3 seconds, then stop squeezing and bring the bottle upright again.
- Lift the bottle clear of the rim while it is upright, carry it back to its marked spot, lower it until it is standing on the tabletop, and release.

One pour, one count. Do not pour a second time, do not shake the bottle, and do not tap it on the rim.

Check: the bottle stands upright on its marked spot, dressing has landed on the salad rather than on the tabletop or down the outside of the mixing bowl, and the mixing bowl is still on its ring.

Step 4: Toss the salad

Goal: the dressing and the toppings are worked through the greens and both servers are back in their rest.

4.1 Take the servers

- The **left gripper** takes the left server from the rest and lifts it straight up out of the slot.
- The **right gripper** takes the right server from the rest and lifts it straight up out of the slot.

4.2 Toss

- Bring both servers down into the mixing bowl on opposite sides, the left server on the far side and the right server on the near side, and push them down to the bottom of the mixing bowl.
- Lift both servers up and over toward the middle of the mixing bowl at the same time, turning what is at the bottom over onto the top.
- Lower them back down the sides and repeat. Make four lift and turn passes in all.

- Keep both servers inside the mixing bowl for the whole toss. Lift them clear only when the four passes are done.

Lift and turn, do not stir. Do not drag the servers in circles around the mixing bowl, do not press the salad down, and do not scrape the sides.

4.3 Return the servers

- Each gripper carries its own server back to the rest, lowers it into its slot until seated, and releases.

Check: the toppings are spread through the greens rather than sitting in a layer on top, nothing was thrown out of the mixing bowl during a pass, the mixing bowl is still on its ring, and both servers are seated in their slots.

Step 5: Portion onto the two plates

Goal: both plates hold a portion and the mixing bowl is left near empty.

Work one plate at a time, taking the **near plate first**.

5.1 Take the servers again

- The **left gripper** takes the left server and the **right gripper** takes the right server, each straight up out of its slot.

5.2 Portion the near plate

- Bring both servers down into the mixing bowl and close them on a portion, the left server underneath and the right server on top, so the portion is held between them.
- Take about half of what is in the mixing bowl on this scoop, judged by eye.
- Lift the portion straight up and clear of the rim, carry it level to the **near plate**, and lower it until it is just above the plate.
- Open both servers together to release the portion onto the middle of the plate, then lift the servers clear.

5.3 Portion the far plate

- Go back to the mixing bowl and take what remains the same way.
- Carry it level to the **far plate** and release it onto the middle of the plate the same way.
- Leave whatever will not come up on the servers in the mixing bowl. Do not scrape the mixing bowl out.

5.4 Return the servers

- Each gripper carries its own server back to the rest, lowers it into its slot until seated, and releases.

Check: both plates hold a portion, the two portions look about the same size by eye, nothing overhangs a plate rim, the mixing bowl is near empty, and both servers are seated in their slots.

Step 6: End the episode

- Confirm the tray holds six empty containers, the bottle stands upright on its spot, the mixing bowl is near empty on its ring, and the tongs and both servers are seated in the rest.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Clear both plates and set the salad aside for disposal.
2. Empty the mixing bowl, wash and dry it, and set it back on its ring in the middle of the tabletop.
3. Wash and dry the tongs and both servers and stand them back in their own slots, handles up.
4. Wipe the bottle spout clean, top the bottle up if it no longer holds enough for a full pour, and stand it upright on its marked spot with the cap off the tabletop.
5. Refill the greens tub with one episode's worth of loose leaves, filled to below the rim.
6. Refill all five topping bowls, one topping each, none above the rim. Vary which topping goes in which topping bowl between episodes rather than rebuilding the same tray every time.
7. Replace any ingredient that is wilted, dried out, or clumped together.
8. Wipe the tabletop and put both plates back empty against the right edge, side by side and not touching.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Filled out of order

- **Visible cue:** a topping goes in before the greens, or the topping bowls are worked in any order other than 1, 2, 3, 4, 5, such as reaching for the nearest full one or skipping ahead and coming back.
- **SOP rule broken:** Steps 1 and 2, tray order, not reach order: greens first, then topping bowl 1 through 5 in that order.
- **Coaching note:** greens first, then straight down the row.

Violation: Ingredient left behind

- **Visible cue:** the episode moves on with the greens tub or a topping bowl still holding some of its contents, or a topping bowl is never visited at all.
- **SOP rule broken:** Steps 1 and 2.1 (take loads until the greens tub is empty; empty each topping bowl, taking a second load with the same tongs if anything is left).
- **Coaching note:** look in the container before you put the tongs down.

Violation: Two tools held at once

- **Visible cue:** a gripper picks up the bottle or a server while the other gripper still holds the tongs, or a server is taken before both hands are empty.
- **SOP rule broken:** Steps 1 to 5, a tool goes back to its rest before the next tool is picked up, and never hold two tools at once.
- **Coaching note:** put it back before you pick the next one up.

Violation: Tool laid down outside its rest

- **Visible cue:** the tongs, the bottle, or a server is set on the tabletop, on the tray, across the bowl rim, or on a plate instead of going back to its slot or, for the bottle, its marked spot.
- **SOP rule broken:** Steps 1, 2.1, 3, 4.3, and 5.4, every tool goes back to its rest between jobs and the bottle goes back to its marked spot.
- **Coaching note:** every tool has one home. It goes there every time.

Violation: Tool regripped in the air

- **Visible cue:** a gripper shifts its hold along the tongs, the bottle, or a server without setting the tool down first.
- **SOP rule broken:** Steps 1 to 5, change jobs through the rest, never in the air.
- **Coaching note:** set it down and take it again.

Violation: Wrong arm on a tool or a zone

- **Visible cue:** the right gripper takes the tongs or picks something off the tray, or the left gripper takes the bottle or reaches a plate while not holding its server.

- **SOP rule broken:** Steps 1 to 5, the left gripper owns the tray and the tongs, the right gripper owns the bottle and the plates, and the left arm reaches the plates only while carrying its server during plating.
- **Coaching note:** left arm loads, right arm dresses and plates. Servers are the only shared job.

Violation: Tool passed hand to hand

- **Visible cue:** the tongs, the bottle, or a server is transferred directly from one gripper to the other in mid air instead of going back to its slot first.
- **SOP rule broken:** Steps 1 to 5, nothing is passed hand to hand, and a tool that changes arms does so by going back to its slot in the rest first.
- **Coaching note:** the rest is the handoff. Nothing changes hands in the air.

Violation: More than one ingredient taken in a trip

- **Visible cue:** the tongs take from two containers before releasing over the mixing bowl, or two toppings are carried across in the same load.
- **SOP rule broken:** Steps 1 and 2, one ingredient at a time, never take from two containers on one trip and never combine two toppings in the tongs.
- **Coaching note:** one container per trip, every trip.

Violation: Ingredient gripped directly

- **Visible cue:** a bare gripper closes on leaves or a topping and moves it, or a gripper is used to push contents into the mixing bowl or off the tongs.
- **SOP rule broken:** Steps 1 to 5, never grip an ingredient directly, everything that goes in the mixing bowl gets there through the tongs or the servers.
- **Coaching note:** the tools touch the food. The grippers never do.

Violation: Load released from height

- **Visible cue:** the tongs open above the mixing bowl rim and the load falls the rest of the way, bouncing or scattering, instead of being lowered inside the rim first.
- **SOP rule broken:** Steps 1 and 2.1, bring the tongs down inside the rim of the mixing bowl before opening them.
- **Coaching note:** get inside the mixing bowl before you open.

Violation: Ingredient dragged over a container rim

- **Visible cue:** the tongs close above a container and scrape or drag the load over its side, or contents are pulled out over the rim rather than lifted straight up.
- **SOP rule broken:** Step 2.1, bring the tongs down inside the rim before closing them, then lift straight up.

- **Coaching note:** close inside, lift straight up.

Violation: Mixing bowl gripped, steadied, or moved

- **Visible cue:** a gripper closes on the mixing bowl or rests against it to hold it still, or the mixing bowl is lifted, slid, or turned at any point during the episode.
- **SOP rule broken:** Steps 1 to 5, never grip, steady, lift, or slide the mixing bowl, the ring holds it.
- **Coaching note:** the ring does that job. If the mixing bowl is moving, stop and reseal it.

Violation: Wrong pour

- **Visible cue:** the bottle is squeezed for clearly longer or shorter than a 3 second count, poured a second time after being brought upright, shaken, tapped on the rim, or tipped before the spout is over the mixing bowl.
- **SOP rule broken:** Step 3 (bring the spout over the middle of the mixing bowl inside the rim, squeeze steadily for a count of 3 seconds, then bring the bottle upright; one pour, one count).
- **Coaching note:** spout over the middle, count three, upright, done.

Violation: Bottle carried or set down tipped

- **Visible cue:** the bottle is lifted away from the mixing bowl still tilted, dripping dressing on the tabletop, or it is put back on its spot leaning or on its side.
- **SOP rule broken:** Step 3 (bring the bottle upright before lifting it clear of the rim, and lower it until it is standing on its marked spot).
- **Coaching note:** upright before you move, upright when you let go.

Violation: Toss is a stir

- **Visible cue:** the servers are dragged in circles around the mixing bowl, pushed through the salad sideways, or used to press the salad down or scrape the sides, instead of going to the bottom and lifting up and over toward the middle.
- **SOP rule broken:** Step 4.2 (push both servers to the bottom on opposite sides and lift up and over toward the middle; lift and turn, do not stir, press, or scrape).
- **Coaching note:** go under it and turn it over. Never push it around.

Violation: Wrong number of toss passes

- **Visible cue:** the servers come out of the mixing bowl after fewer than four lift and turn passes, or the tossing continues past four.
- **SOP rule broken:** Step 4.2 (make four lift and turn passes in all).
- **Coaching note:** count the passes out loud. Four, then out.

Violation: Servers lifted out mid toss

- **Visible cue:** one or both servers come clear of the mixing bowl between passes, or one server is parked outside the mixing bowl while the other works alone.
- **SOP rule broken:** Step 4.2 (keep both servers inside the mixing bowl for the whole toss and lift them clear only when the four passes are done).
- **Coaching note:** both servers stay in until the count is done.

Violation: Wrong plating order

- **Visible cue:** the far plate is filled before the near plate, or both portions go onto one plate.
- **SOP rule broken:** Step 5 (work one plate at a time, taking the near plate first).
- **Coaching note:** near plate first, one portion each.

Violation: Uneven portions

- **Visible cue:** the first scoop clearly takes most or nearly none of what is in the mixing bowl, leaving the two plates visibly different in size.
- **SOP rule broken:** Step 5.2 (take about half of what is in the mixing bowl on the first scoop, judged by eye).
- **Coaching note:** look at the mixing bowl before you close the servers. Half, then the rest.

Violation: Bowl scraped out

- **Visible cue:** the servers are dragged around the inside of the mixing bowl or tipped against its wall to work the last of the salad out after the far plate is filled.
- **SOP rule broken:** Step 5.3 (leave whatever will not come up on the servers in the mixing bowl; do not scrape the mixing bowl out).
- **Coaching note:** what stays in the mixing bowl stays in the mixing bowl.

Violation: Portion dropped instead of lowered

- **Visible cue:** the servers open above a plate and the portion falls the rest of the way, scattering across the plate or off its rim, instead of being lowered to just above the plate first.
- **SOP rule broken:** Step 5.2 and Step 5.3 (lower the portion until it is just above the plate, then open both servers together).
- **Coaching note:** take it all the way down before you open.

Violation: Salad pressed or packed

- **Visible cue:** a server or a gripper presses down on the contents of the mixing bowl or on a plated portion to flatten or shape it.
- **SOP rule broken:** Steps 4.2 and 5, never press or pack the salad down.
- **Coaching note:** leave it how it lands.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the tray, over a topping bowl, or over open tabletop, or a tool is released anywhere other than its slot or, for the bottle, its marked spot.
- **SOP rule broken:** Steps 1 to 5, a gripper opens only over the mixing bowl, over a plate, or into a slot in the rest.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Spill collected mid episode

- **Visible cue:** an arm stops to pick up, sweep, or push leaves or toppings that have landed on the tabletop back into the mixing bowl or a container.
- **SOP rule broken:** Steps 1 to 5, spilled leaves or toppings on the tabletop are left where they fall.
- **Coaching note:** keep going. Spills are cleared in the reset, not on camera.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** a topping bowl or the greens tub is tipped or dragged across the tray, the bottle is knocked over, the mixing bowl is struck off its ring, a plate is pushed out of its spot, or a tool strikes the mixing bowl rim hard enough to move it.
- **SOP rule broken:** Steps 1 to 5, carry every tool level and clear of the rim of anything it passes over, and release only after a stable placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a tool out of its rest, the bottle off its spot or lying down, a container still holding contents, an arm away from home.
- **SOP rule broken:** Step 6 (confirm the empty containers, the upright bottle, the near empty mixing bowl, and all three tools seated; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective ingredient:** greens found wilted or frozen together, or a topping found clumped into a mass the tongs cannot separate. Replace it before the next episode.
- **Dressing bottle runs empty or will not pour** under a steady squeeze, leaving the salad undressed. Top it up or clear the spout before the next episode.

- **Tool or rest fault:** tongs that will not close on a load, a server that comes loose in the gripper, or a rest slot that does not hold a tool upright.
- **Mixing bowl leaves its ring** under its own weight, or the ring slides on the tabletop, so the mixing bowl cannot be worked without being held. Reseat both before the next episode.
- **A plate or the tray shifts out of its zone** during the episode.

Annotation subtasks (from SOP)

1. Take the tongs from the rest
2. Move one load of greens from the tub to the mixing bowl
3. Move one load of a topping from a topping bowl to the mixing bowl
4. Return the tongs to the rest
5. Take the dressing bottle from its spot
6. Pour dressing over the mixing bowl
7. Return the bottle to its spot
8. Take both servers from the rest
9. Toss the salad with four lift and turn passes
10. Return both servers to the rest
11. Move one portion from the mixing bowl to a plate
12. Return both arms home and end the episode